

Head-up Display Based on Micro-lens to Improve Micro-LED Brightness

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Abstract

We have studied the technology of preparing micro-lens on micro-LED display. Through appropriate pixel arrangement and the design of each sub-pixel corresponding to each micro-lens, we can narrow the near-Lambertian light distribution of micro-LED and make the maximum brightness of the panel four times that without micro-lens. Then, according to the misalignment relationship between the center positions of the micro-lens and micro-LED, the main viewing angle directions of different positions on a display are different to meet the main viewing angle requirements of AR-HUD for the panel. Finally, based on the optical design, an AR-HUD with a large FOV of about $15.4^{\circ} \times 7.7^{\circ}$ was prepared. The peak eye brightness is $>18000\text{nit}$, the NTSC exceeds 110%, and the eye box brightness uniformity is $>70\%$. It fully meets the viewing angle required by the optical and mechanical design, which greatly promotes the application scope of Micro-LED.

Author Keywords

Micro-LED, Micro-lens, heads up display(HUD), FOV

1. Introduction

With the upgrade of smart cockpits, AR-HUD has become the core interactive device to improve driving safety, and the market size is increasing year by year^[1-2]. The application of large-scale AR-HUD will bring about technological updates and iterations. The display effect of AR-HUD is obviously restricted by light source technology. Current mainstream light sources such as LCD have insufficient brightness, DLP has a complex structure, and laser costs are high, making it difficult to balance strong light adaptability, miniaturization, and energy consumption requirements. Micro-LED technology provides a solution to this dilemma^[3].

First of all, micro-LED can achieve very high brightness, can resist sunlight interference, and has a high contrast, which effectively solves the problem of graying of the display. Secondly, micro-LED light source has a high color gamut, NTSC $>110\%$. During driving, the high-color gamut warning icon can attract the driver's attention and provide safety guarantee. Finally, the long life of the micro-LED display, wider operating temperature range and better stability^[3-4] are undoubtedly another better choice for vehicle display. As the industry continues to upgrade to high brightness, high safety, high stability, wide operating temperature range, and full-scenario adaptability, Micro-LED has become a key direction for breakthroughs in AR-HUD technology.

This research is based on LTPS glass-based Micro-LED. It mainly completes the substrate pixel design and optical structure design to increase the display brightness by 3 times, and the main viewing angles are different at different panel positions. Through process and process development such as driver debugging and opto-mechanical design, an AR-HUD device using glass-based micro-LED as the image source was successfully prepared. Its FOV is as high as $15.4^{\circ} \times 7.7^{\circ}$, NTSC $>110\%$, and eye brightness $>18000\text{nit}$.

2. Simulation design

2.1 Opto-mechanical design and its requirements for Micro-LED display

In the AR-HUD optical-mechanical design, a two-piece reflector design is used, simulated and optimized with zemax optical simulation software, to accurately match the eye box with a range of $130\text{mm} \times 50\text{mm}$, and optimize the curvature, spacing and other parameters of the reflector, resulting in an optical-mechanical system with good imaging quality, with distortion controlled within 3%. The light requirements for Micro-LED are shown in Figure 1 and Table 1: ① The panel is in the H direction, and as the position of the panel changes, the main viewing angle gradually changes; the same is true in the V direction; ② The maximum required viewing angle width is about 23° .

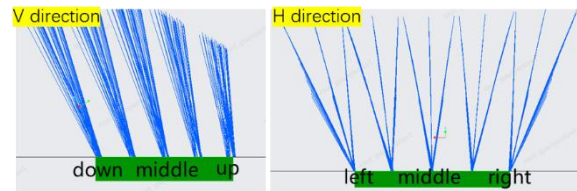


Figure 1. (a) Schematic diagram of the viewing angle requirements in the H direction (b) Schematic diagram of the viewing angle requirements in the V direction

Table 1. Main viewing angle and viewing angle width at different positions of the panel

H direction (left is negative)	left	middle	right
Main Angle/Angle of View Width	$-16^{\circ}/23^{\circ}$	$-1.5^{\circ}/19^{\circ}$	$13^{\circ}/21^{\circ}$
V direction	up	middle	down
Main Angle/Angle of View Width	$10^{\circ}/7^{\circ}$	$15^{\circ}/7^{\circ}$	$22^{\circ}/7^{\circ}$

2.2 Micro-LED display test and problem:

1、Micro-LED has high luminous efficiency, but because Micro-LED is a volume light source, the light distribution is relatively divergent, similar to a Lambert body, as shown in Figure 2 (blue is the H direction, green is the V direction). In the design of AR-HUD, the brightness of the panel is very high, but the viewing angle width of the panel is not high. It only needs about 23° to meet the viewing needs. Therefore, as the display light source of AR-HUD, Micro-LED has a light distribution that is too wide, which will cause a lot of waste of light energy;

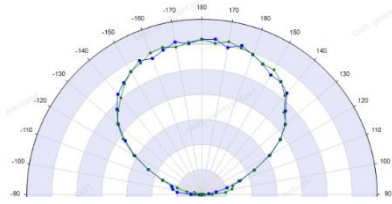


Figure 2. Cutaway view of Micro-LED light emission angle

2、The viewing angle of the Micro-LED panel will not change with the change of different positions of the panel. When tested at different panel positions, the viewing angle remains basically the same. However, according to the simulation design of AR-HUD, the requirements for the panel are that the main viewing angle requirements are different at different panel positions. Although this can be solved by a Fresnel lens, adding a Fresnel lens will not only increase the cost, but also cause a loss of brightness due to the transmittance of the Fresnel lens;

2.3 simulation design

The glass-based Micro-LED display studied in this article has a size of 4.6 inches, a resolution of 1280*720, and a pitch of 80μm. Micro-LED chip sizes are available in a variety of options, including 15μm×25μm(MQW size 15μm×13μm), 15μm×30μm (MQW size 23μm × 13μm), 25μm×30μm (MQW size 23μm×18μm).

Optical narrowing can improve brightness. Theoretical calculations are as follows:

$$2\pi l \int_0^{\frac{\pi}{2}} \cos \theta \sin \theta d\theta = 2\pi X \int_0^{\frac{y}{2}} \cos \theta \sin \theta d\theta$$

l is the brightness, X is the brightness after light shaping, y is the narrowed viewing angle width, considering that the alignment accuracy y is 30° (in line with the half-width of the viewing angle of AR-HUD), $X / l \approx 14.9$ is calculated. That is to say, if all the light energy is received at available angles, the brightness will be increased by at least 14 times. However, Micro-LED is small in size and it is almost impossible to use reflective cups or TIR lenses to collect light. Therefore, no matter what the design, a large part of the light energy will be wasted.

We use Micro-lens, which has low brightness improvement and feasible process, as the structure to gather light energy. According to analysis, the larger the radius of the Micro-lens, the higher the brightness improvement, but there are two limitations: 1. The pitch size of the panel. The size of the Micro-lens is limited by the panel pitch size and cannot be infinitely large; 2. AR-HUD requires the viewing angle width of the display module.

According to analysis, making the lens as large as possible under 80μm pitch will still not be lower than the required viewing angle width. Our glyph pixel arrangement design can maximize the diameter of the Micro-lens and improve the utilization of Micro-LED light. The design radius of the Micro-lens is 22μm, the thickness is 17.5μm, and the distance between the lens and the LED is about 30μm, as shown in Figure 3(a).

For the selection of chip size, we conducted simulation research. It is found that after fixing the size of the Micro-lens, the brightness improvement is mainly related to the effective light-emitting area of the chip, that is, to the MQW area. The smaller the area, the higher the brightness improvement, as shown

in Figure 3(b), but the corresponding viewing angle width is narrower. According to this analysis, the light-emitting area of the RGB three-color chip must be consistent, because if the RGB light-emitting area is inconsistent, the viewing angle width will be different, which will cause serious color shift. So we choose 15μm×25μm chip.

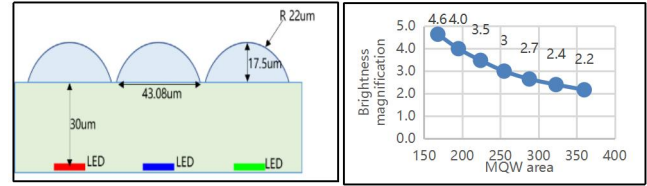


Figure 3. (a) Micro-lens design diagram (b) Relationship between brightness improvement and MQW

The Micro-lens simulation results show that, as shown in Figure 4 (blue is the H direction, green is the V direction), the brightness is about 4 times the brightness without Micro-lens;

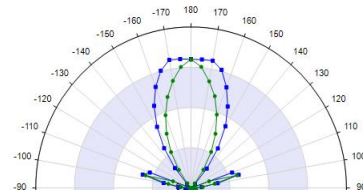


Figure 4. Cutaway view of light distribution after adding Micro-lens

After encapsulation without micro-lens, the light distribution is close to Lambertian. As shown in Figure 2, the visible light distribution is divergent. This distribution causes a great waste of light energy. After adding micro-lens, the emitted light distribution is greatly narrowed (Figure 4). The main viewing angle light energy is enhanced, and the brightness is about 4 times the original. And its half viewing angle width is narrowed to about 30° in the H direction, and the viewing angle in the V direction is narrowed to about 25°.

The design of Micro-Lens basically confirmed the brightness improvement through simulation. Then we confirmed the design method of different main viewing angles at different positions of the panel based on the positional relationship between Micro-lens and Micro-LED;

The basic principle is to confirm the main viewing angle of the position based on the position offset of the Micro-lens and Micro-LED, as shown in Figure 5(a) below. This figure is a schematic diagram of the principle of simulating different panel positions to have different viewing angles. we design different viewing angles at different positions of the panel based on the offset of the center positions of the Micro-lens and Micro-LED. Through simulation, the viewing angles of the left, middle and right positions in the H direction of the panel and the upper, middle and lower positions in the V direction are simulated respectively, as shown in Figure 5(b), to confirm their offset positions, and then design the pitch of the micro-lens based on calculation (which is slightly different from the pitch of the panel pixels) to achieve the purpose of continuous changes in the viewing angles at different positions on the panel;

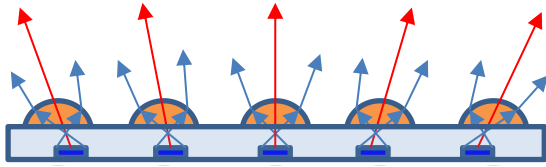


Figure 5(a). Design diagram of different viewing angles at different panel positions

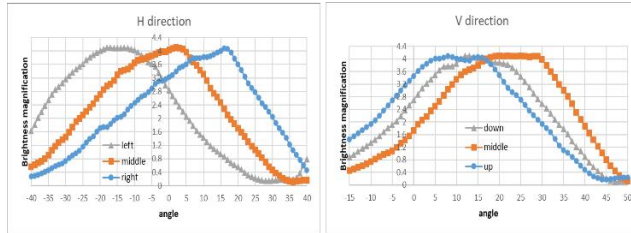


Figure 5(b). Simulation magnification data of viewing angles at different positions on the panel

3. results

3.1 Micro-lens preparation:

We prepared micro-lens by imprinting based on the 4.63-inch panel. The difference between the actual effect and the design is very small. The diameter of the actual test lens is about 42μm and the radius is about 22μm. According to the actual test, the effect of the actual preparation of the micro-lens is in line with the design;

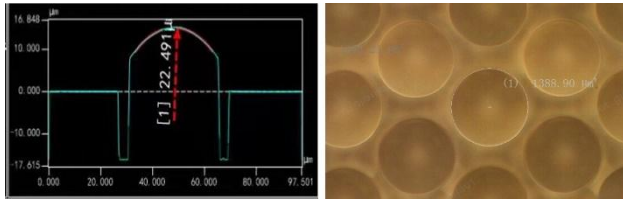


Figure 6. (a) Lens radius test (b) Lens caliber microscope test chart

3.2 Analysis of brightness improvement and viewing angle test results:

3.2.1 The actual test brightness improvement data is shown in Figure 7 below:

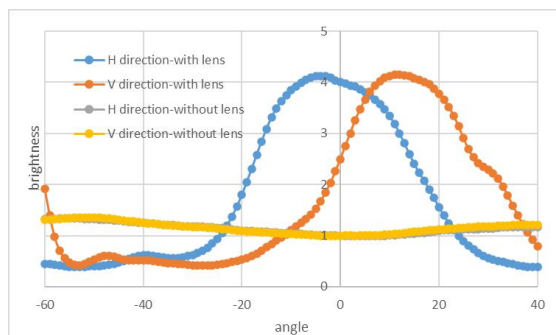


Figure 7. Brightness changes before and after micro-lens

This test point is the center point of the panel. The main viewing angle in the H direction is about -1.5° , and the main viewing angle in the V direction is about 15° , which is consistent with the main viewing angle design. After adding micro-lens, the performance is about 4 times that without micro-lens, which is very different from the simulation estimate. It can be seen from the figure that the brightness without micro-lens changes very little with the angle. After adding micro-lens, the light energy in the H direction is mainly concentrated between $-17^\circ \sim 14^\circ$, and the viewing angle width is about 31° . The light energy in the V direction is mainly concentrated between $2^\circ \sim 27^\circ$, and the viewing angle width is about 25° . The actual viewing angle width exceeds the required 23° in the H direction and 7° in the V direction. This design is actually to prevent alignment deviation when preparing the lens, so the viewing angle range is intentionally expanded.

The simulation and actual brightness improvements are comparable, but there are still some deviations in the overlap of the light distribution curves. The main reason is that the surface brightness of the micro-LED is uneven.

3.2.2 The simulated brightness and viewing angle width have been experimentally verified, and the test of the main viewing angle at different positions of the panel also reflects the success of the overall panel micro-lens design. We tested three points from left to right according to the H direction of the panel. The left, midpoint and right sides, the main viewing angles are -16° , -1.5° and 13° respectively, that is, the H direction should gradually increase from negative angle to positive angle. The actual test is shown in Figure 8 below. It can be seen that the angle well illustrates the different main viewing angles at different panel positions, and the main viewing angle position is consistent with the actual simulation design.

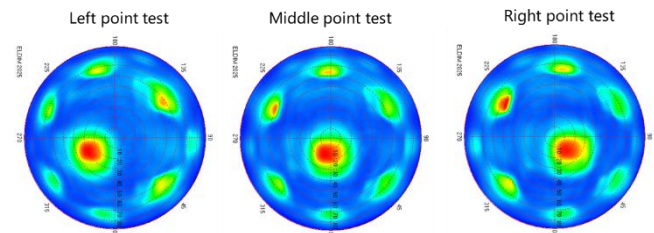


Figure 8. Test cloud images at different positions in the H direction

3.4 Optical system and display module:

This 4.63-inch (1280*720 pixels) full-color display module uses a TFT glass-based substrate to integrate a Micro-LED array. It is specially designed for automotive head-up display (AR-HUD) applications. It runs at a 120Hz refresh rate and has a peak brightness of up to 140,000nit, ensuring visibility in strong light environments. Its color gamut reaches 110% (the traditional display module is about 70% of the AR-HUD color gamut of LCD), which enriches the display colors of AR-HUD. Its high-purity colors can better catch the eye when displaying navigation and display warnings, adding a solid guarantee for driving safety. Micro-LED's pixel-level light emission and 0 grayscale almost black display mode increase display contrast;

Opto-mechanical design achieves a super large FOV of $15.4^\circ \times 7.7^\circ$, a downward viewing angle of about 2° , a contrast

ratio of over 100,000:1, and a virtual image design of about 10m to better integrate with the road. With the help of Micro-LED ultra-high brightness and high color gamut display module, core information such as navigation and vehicle speed can be displayed in real time, reducing the need for drivers to deviate from the road. By projecting key data into the driver's line of sight, it improves the driver's environmental perception and supports more accurate judgments of road conditions, thereby potentially improving driving safety.

Table 2. AR-HUD specific parameters

	AR-HUD parameter
resolution	1280*720
PPI	317
Eye brightness	>18000nit
Contrast	100000: 1
Color gamut /NTSC	112%
downward perspective	2°
FOV(H*V)	15.4°*7.7°
exit pupil distance	800mm
Eyebox (H*V)	130mm*50mm
virtual image distance	10±1m



Figure 9 (a). Display module information



Figure 9(b). Optical system structure diagram

4. Conclusion

In this study, by designing micro-lens, the near-Lambertian light distribution of micro-LED is narrowed, achieving an effect of increasing the brightness by about 3 times. And through the deviation of the center alignment of the micro-lens and micro-LED, different main viewing angles at different positions of the panel are achieved to meet the AR-HUD's requirements for panel viewing angles. Optical simulation is used to design the specific parameters of the micro-lens, and estimate the brightness improvement and main viewing angle shift, and then verify the simulation based on actual preparation.

The design and preparation of Micro-lens has successfully improved the brightness of Micro-LED displays by about 4 times. Using this technology, we successfully developed a glass-based 4.6-inch (1280*720 pixels) AR-HUD display module with ultra-high brightness and 110% color gamut. Based on this high-brightness and high color gamut display module, we further designed and developed an AR-HUD opto-mechanical system, providing more valuable experience for glass-based micro-LED applications in AR-HUD.

5. References

1. Pautz, A. (2015). Head Up Display in Automotive: A New Reality for the Driver. Cham: Springer International Publishing. p. 505-516.
2. Gabbard, J.L., G.M. Fitch and H. Kim, (2014). Behind the Glass: Driver Challenges and Opportunities for AR Automotive Applications. Proceedings of the IEEE. 102(2), p. 124-136. doi:10.1109/JPROC.2013.2294642
3. Liu M, Nie J, Liu Y, et al. Research on the reliability of Micro LED high-density solder joints under thermal cycling conditions[J]. IOP Publishing Ltd, 2022.
4. Yang Q, Yang Z, Lan Y F, et al. Low-diffraction transparent micro light-emitting diode displays with optimized pixel structure[J]. Journal of the Society for Information Display, 2022(4/6):30